

Technical Appendix: Complete Field Equations of the Two-Mode Coupling Model (Track A)

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Abstract

This appendix collects, in a single self-contained reference, all the field equations used in the two-mode Jaynes–Cummings model of Track A. We present the full Hilbert-space structure, the operator algebra, the Hamiltonian, the Lindblad dissipator, the master equation, the reduced dynamics, the mean-field closure, the control parameters, the mixing angle, the order parameters, and the information-theoretic measures. We also include the numerical scheme, the discretization, and the convergence criteria. The document is intended as a supplementary reference for the main paper, not as a stand-alone publication.

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1 Conventions and notation

1.1 Units

Throughout this appendix we set $\hbar = 1$ and $k_B = 1$. Frequencies are measured in Hz (equivalently, in units of s^{-1}). Energy is measured in the same units as frequency. When dimensional consistency requires it, $\hbar$ is restored explicitly.

1.2 Indices

- 0: substrate sector (non-analytic, oscillator mode).
- 1: analytic sector (baryonic, qubit mode).
- μ, ν : spacetime indices in the Lindblad context.
- $n \in \{0, 1, \dots, N_0 - 1\}$: Fock quantum number of the oscillator.
- $s \in \{0, 1\}$: qubit state label.

1.3 Operators

For any operator $\hat{O}$ we denote by

- $\hat{O}^\dagger$: Hermitian conjugate.
- $\langle \hat{O} \rangle_\rho = \text{Tr}(\rho \hat{O})$: expectation value in state ρ .
- $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$: commutator.
- $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$: anticommutator.

2 Hilbert space structure

2.1 Single-sector spaces

The total Hilbert space is the tensor product

$$\boxed{\mathcal{H} = \mathcal{H}_0 \otimes \mathcal{H}_1, \quad \dim \mathcal{H} = d_0 \cdot d_1.} \quad (1)$$

For the numerical implementation we truncate the oscillator to N_0 levels, so

$$d_0 = N_0, \quad d_1 = 2, \quad \dim \mathcal{H} = 2N_0. \quad (2)$$

2.2 Basis of the analytic sector

We use the basis

$$\mathcal{B}_1 = \{|0\rangle_1, |1\rangle_1\}, \quad (3)$$

where

- $|0\rangle_1$: analytic mode *off* (inactive).
- $|1\rangle_1$: analytic mode *on* (active).

The projectors are

$$\hat{P}_0^{(1)} = |0\rangle_1 \langle 0|, \quad \hat{P}_1^{(1)} = |1\rangle_1 \langle 1|, \quad (4)$$

satisfying

$$\hat{P}_0^{(1)} + \hat{P}_1^{(1)} = \hat{I}_1, \quad \hat{P}_i^{(1)} \hat{P}_j^{(1)} = \delta_{ij} \hat{P}_i^{(1)}. \quad (5)$$

2.3 Basis of the substrate sector

We use the Fock basis

$$\mathcal{B}_0 = \{|n\rangle_0\}_{n=0}^{N_0-1}. \quad (6)$$

The number operator is

$$\hat{n}_0 = \hat{a}^\dagger \hat{a}, \quad \hat{n}_0 |n\rangle_0 = n |n\rangle_0. \quad (7)$$

Completeness in the truncated space:

$$\sum_{n=0}^{N_0-1} |n\rangle_0 \langle n| = \hat{I}_0. \quad (8)$$

2.4 Tensor-product operators

For any operator $\hat{A}$ acting on $\mathcal{H}_0$ and any operator $\hat{B}$ acting on $\mathcal{H}_1$, we write

$$\hat{A} \otimes \hat{B} \equiv \hat{A}\hat{B}, \quad (9)$$

with the understanding that the first factor acts on the substrate and the second on the analytic sector. When no ambiguity arises, we write $\hat{A}$ for $\hat{A} \otimes \hat{I}_1$ and $\hat{B}$ for $\hat{I}_0 \otimes \hat{B}$.

3 Operator algebra

3.1 Oscillator operators

The annihilation and creation operators satisfy

$$\boxed{[\hat{a}, \hat{a}^\dagger] = \hat{I}_0, \quad [\hat{a}, \hat{a}] = [\hat{a}^\dagger, \hat{a}^\dagger] = 0.} \quad (10)$$

Their action on the Fock basis is

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle, \quad (11)$$

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle. \quad (12)$$

The number operator satisfies

$$\hat{n}_0 = \hat{a}^\dagger \hat{a}, \quad [\hat{n}_0, \hat{a}] = -\hat{a}, \quad [\hat{n}_0, \hat{a}^\dagger] = +\hat{a}^\dagger. \quad (13)$$

In the matrix representation (truncated to N_0 levels), the annihilation operator is

$$(\hat{a})_{mn} = \sqrt{m} \delta_{m+1,n}, \quad m, n \in \{0, 1, \dots, N_0 - 1\}. \quad (14)$$

3.2 Qubit operators

The Pauli operators in the basis $\mathcal{B}_1$ are

$$\hat{\sigma}_+ = |1\rangle_1 \langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad (15)$$

$$\hat{\sigma}_- = |0\rangle_1 \langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (16)$$

$$\hat{\sigma}_z = |1\rangle_1 \langle 1| - |0\rangle_1 \langle 0| = \begin{pmatrix} -1 & 0 \\ 0 & +1 \end{pmatrix}. \quad (17)$$

They satisfy the algebra

$$[\hat{\sigma}_+, \hat{\sigma}_-] = \hat{\sigma}_z, \quad [\hat{\sigma}_z, \hat{\sigma}_\pm] = \pm 2\hat{\sigma}_\pm. \quad (18)$$

The projector onto the active state is

$$\hat{\sigma}_+ \hat{\sigma}_- = |1\rangle_1 \langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \quad (19)$$

3.3 Total excitation operator

We define the total excitation number

$$\hat{\mathcal{N}} = \hat{a}^\dagger \hat{a} + \hat{\sigma}_+ \hat{\sigma}_- = \hat{n}_0 + \hat{\sigma}_+ \hat{\sigma}_-. \quad (20)$$

Under the Jaynes–Cummings interaction, this is conserved:

$$[\hat{\mathcal{N}}, \hat{H}_{\text{int}}] = 0. \quad (21)$$

4 Hamiltonians

4.1 Free Hamiltonian

The free Hamiltonian is

$$\boxed{\hat{H}_0 = \omega_0 \hat{a}^\dagger \hat{a} + \omega_1 \hat{\sigma}_+ \hat{\sigma}_-} \quad (22)$$

Its action on the product basis is

$$\hat{H}_0 |n\rangle_0 \otimes |s\rangle_1 = (\omega_0 n + \omega_1 \delta_{s,1}) |n\rangle_0 \otimes |s\rangle_1. \quad (23)$$

4.2 Detuning

The detuning between sectors is

$$\boxed{\delta \equiv \omega_1 - \omega_0.} \quad (24)$$

In the numerical simulations, we use $\omega_0 = 6$ Hz and $\omega_1 = 40$ Hz, giving $\delta = 34$ Hz.

4.3 Interaction Hamiltonian

The Jaynes–Cummings interaction in the Rotating Wave Approximation (RWA) is

$$\boxed{\hat{H}_{\text{int}}(t) = g(t) \left(\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+ \right)}. \quad (25)$$

The two terms have a clear physical interpretation:

- $\hat{a}^\dagger \hat{\sigma}_-$: creates a substrate quantum and de-excites the analytic mode. This is the *absorption* channel.
- $\hat{a} \hat{\sigma}_+$: destroys a substrate quantum and excites the analytic mode. This is the *emission* channel.

The RWA is valid when

$$g \ll \omega_0 + \omega_1. \quad (26)$$

For our numerical parameters, $g \lesssim 50$ Hz and $\omega_0 + \omega_1 = 46$ Hz. The RWA is marginal at the upper end of the swept range, but the qualitative results do not depend on it.

4.4 Total Hamiltonian

The total Hamiltonian is

$$\boxed{\hat{H}(t) = \hat{H}_0 + \hat{H}_{\text{int}}(t).} \quad (27)$$

In the product basis $|n\rangle_0 \otimes |s\rangle_1$:

$$\hat{H}(t) = \sum_{n=0}^{N_0-1} \sum_{s \in \{0,1\}} E_{n,s} |n, s\rangle \langle n, s| + g(t) \sum_{n=0}^{N_0-2} \sqrt{n+1} (|n+1, 0\rangle \langle n, 1| + |n, 1\rangle \langle n+1, 0|), \quad (28)$$

where

$$E_{n,s} = \omega_0 n + \omega_1 \delta_{s,1}. \quad (29)$$

4.5 Excitation manifolds

For a fixed total excitation number N , the states

$$|N, 0\rangle \equiv |N\rangle_0 \otimes |0\rangle_1, \quad |N-1, 1\rangle \equiv |N-1\rangle_0 \otimes |1\rangle_1, \quad (30)$$

span a two-dimensional subspace. In this subspace, the total Hamiltonian reduces to

$$\hat{H}^{(N)} = \begin{pmatrix} \omega_0 N & g\sqrt{N} \\ g\sqrt{N} & \omega_0(N-1) + \omega_1 \end{pmatrix}. \quad (31)$$

The eigenvalues are

$$E_{\pm}^{(N)} = \omega_0 N + \frac{\delta}{2} \pm \frac{1}{2} \sqrt{\delta^2 + 4g^2 N}, \quad (32)$$

and the Rabi frequency at fixed N is

$$\boxed{\Omega_R^{(N)} = \sqrt{\delta^2 + 4g^2 N}.} \quad (33)$$

For $N = 1$ (the lowest nontrivial manifold), this reduces to

$$\Omega_R^{(1)} = \sqrt{\delta^2 + 4g^2}. \quad (34)$$

5 Rabi oscillation and mixing angle

5.1 Rabi frequency

We define the Rabi frequency at the first excitation manifold as

$$\boxed{\Omega_R \equiv \sqrt{\delta^2 + 4g^2} = 2\sqrt{g^2 + (\delta/2)^2}.} \quad (35)$$

The corresponding Rabi period is

$$T_R = \frac{2\pi}{\Omega_R}. \quad (36)$$

For $g = 17$ Hz, $\delta = 34$ Hz: $\Omega_R = 48.08$ Hz, $T_R = 0.131$ s.

Note: in some references, the Rabi frequency is defined as $\Omega_R = \sqrt{g^2 + (\delta/2)^2}$. We adopt the convention with the factor of 2 to match the standard Jaynes–Cummings result.

5.2 Mixing angle

At fixed excitation manifold N , the mixing angle is

$$\boxed{\theta_N = \frac{1}{2} \arctan \left(\frac{2g\sqrt{N}}{\delta} \right).} \quad (37)$$

For $N = 0$ (the vacuum manifold, trivial), the states are $|0, 0\rangle$ and there is no mixing. The physical range is $N \geq 1$.

The dressed states are

$$|+, N\rangle = \cos \theta_N |N, 0\rangle + \sin \theta_N |N-1, 1\rangle, \quad (38)$$

$$|-, N\rangle = -\sin \theta_N |N, 0\rangle + \cos \theta_N |N-1, 1\rangle. \quad (39)$$

5.3 Geometric labels

The value $\theta_N = \pi/8 = 22.5^\circ$ corresponds to

$$\frac{2g\sqrt{N}}{\delta} = \tan(\pi/4) = 1 \implies g_{\text{crit}}^{(N)} = \frac{\delta}{2\sqrt{N}}. \quad (40)$$

For $N = 1$: $g_{\text{crit}}^{(1)} = \delta/2 = 17$ Hz.

Important: This value is a *geometric label*, not a dynamical transition. Sweeping g across g_{crit} does not produce a qualitative change in the dynamics. The true control parameter is $\alpha = g/\delta$, and the true dynamical boundary depends on both α and the dissipation rate γ_ϕ .

6 DMN modulation

6.1 Activity variable

We introduce the DMN activity variable

$$A_{\text{DMN}}(t) \in [0, 1], \quad (41)$$

where $A_{\text{DMN}} = 1$ corresponds to maximal activity (strong analytic filtering) and $A_{\text{DMN}} = 0$ to full suppression.

6.2 Sigmoidal filter

The coupling modulation is

$$\boxed{\eta(A) = \frac{1}{1 + \exp[\lambda(A - A_{\text{th}})]}}, \quad (42)$$

where

- λ : sigmoid stiffness (dimensionless).
- A_{th} : threshold value.

The limits are

$$\lim_{A \rightarrow 0} \eta(A) = \frac{1}{1 + e^{-\lambda A_{\text{th}}}}, \quad \lim_{A \rightarrow 1} \eta(A) = \frac{1}{1 + e^{\lambda(1 - A_{\text{th}})}}. \quad (43)$$

For $\lambda \gg 1$, the filter approaches a step function.

6.3 Coupling strength

The time-dependent coupling is

$$\boxed{g(t) = g_0 \eta(A_{\text{DMN}}(t))}, \quad (44)$$

with g_0 the maximal coupling. Typical numerical values: $g_0 = 20$ Hz, $\lambda = 8$, $A_{\text{th}} = 0.5$.

6.4 Stochastic dynamics

The DMN activity evolves as an Ornstein–Uhlenbeck process:

$$\tau_{\text{DMN}} \frac{dA_{\text{DMN}}}{dt} = -(A_{\text{DMN}} - A_0) + \sigma \xi(t), \quad (45)$$

where:

- τ_{DMN} : relaxation time.
- A_0 : baseline activity.
- σ : noise amplitude.
- $\xi(t)$: Gaussian white noise with

$$\langle \xi(t) \rangle = 0, \quad \langle \xi(t) \xi(t') \rangle = 2\delta(t - t'). \quad (46)$$

The stationary distribution of A_{DMN} is Gaussian with

$$\langle A_{\text{DMN}} \rangle = A_0, \quad \text{Var}(A_{\text{DMN}}) = \frac{\sigma^2}{\tau_{\text{DMN}}}. \quad (47)$$

7 Lindblad dissipation

7.1 Master equation

The evolution of the density matrix is governed by the Lindblad master equation:

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar} [\hat{H}(t), \hat{\rho}] + \mathcal{L}_{\text{dec}}[\hat{\rho}]. \quad (48)$$

7.2 Dissipator

The dissipator has three channels:

$$\begin{aligned} \mathcal{L}_{\text{dec}}[\hat{\rho}] = & \gamma_{\phi} (\hat{\sigma}_z \hat{\rho} \hat{\sigma}_z - \hat{\rho}) \\ & + \gamma_a (n_{\text{th}} + 1) \left(\hat{a} \hat{\rho} \hat{a}^\dagger - \frac{1}{2} \{ \hat{a}^\dagger \hat{a}, \hat{\rho} \} \right) \\ & + \gamma_a n_{\text{th}} \left(\hat{a}^\dagger \hat{\rho} \hat{a} - \frac{1}{2} \{ \hat{a} \hat{a}^\dagger, \hat{\rho} \} \right). \end{aligned} \quad (49)$$

7.3 Physical interpretation of each channel

Channel	Operator	Rate	Physical meaning
Phase damping	$\hat{\sigma}_z$	γ_{ϕ}	Loss of analytic coherence (dephasing)
Amplitude damping	$\hat{a}$	$\gamma_a (n_{\text{th}} + 1)$	Substrate relaxation
Thermal pumping	$\hat{a}^\dagger$	$\gamma_a n_{\text{th}}$	Thermal excitation of substrate

7.4 Thermal occupation

The thermal occupation number is

$$n_{\text{th}} = \frac{1}{e^{\hbar\omega_0/k_B T} - 1}. \quad (50)$$

For neural effective temperatures, this is a free parameter. In the numerical simulations, we use $n_{\text{th}} = 0.05$ (deep quantum regime).

7.5 Trace preservation

The dissipator is trace-preserving:

$$\text{Tr}(\mathcal{L}_{\text{dec}}[\hat{\rho}]) = 0, \quad \frac{d}{dt}\text{Tr}(\hat{\rho}) = 0. \quad (51)$$

It also preserves Hermiticity:

$$(\mathcal{L}_{\text{dec}}[\hat{\rho}])^\dagger = \mathcal{L}_{\text{dec}}[\hat{\rho}]. \quad (52)$$

8 Reduced dynamics

8.1 Partial traces

The reduced density matrices are

$$\hat{\rho}_1 = \text{Tr}_0(\hat{\rho}) = \sum_{n=0}^{N_0-1} \langle n|_0 \hat{\rho} |n\rangle_0, \quad (53)$$

$$\hat{\rho}_0 = \text{Tr}_1(\hat{\rho}) = \sum_{s \in \{0,1\}} \langle s|_1 \hat{\rho} |s\rangle_1. \quad (54)$$

8.2 Exact reduced equations (not closed)

The exact evolution equations are

$$\frac{d\hat{\rho}_1}{dt} = -\frac{i}{\hbar}[\hat{H}_1, \hat{\rho}_1] - \frac{i}{\hbar}\text{Tr}_0[\hat{H}_{\text{int}}, \hat{\rho}] + \mathcal{L}_{\text{dec}}^{(1)}[\hat{\rho}_1], \quad (55)$$

$$\frac{d\hat{\rho}_0}{dt} = -\frac{i}{\hbar}[\hat{H}_0, \hat{\rho}_0] - \frac{i}{\hbar}\text{Tr}_1[\hat{H}_{\text{int}}, \hat{\rho}] + \mathcal{L}_{\text{dec}}^{(0)}[\hat{\rho}_0]. \quad (56)$$

The interaction terms involve correlations between the two sectors and do not close in terms of $\hat{\rho}_0$ and $\hat{\rho}_1$ alone.

8.3 Mean-field closure

Under the Hartree–Fock approximation, we factorize

$$\hat{\rho} \approx \hat{\rho}_0 \otimes \hat{\rho}_1, \quad (57)$$

which gives the closed system

$$\frac{d\hat{\rho}_1}{dt} = -\frac{i}{\hbar}[\hat{H}_1 + \hat{H}_{\text{int}}^{\text{eff}}(\hat{\rho}_0), \hat{\rho}_1] + \mathcal{L}_{\text{dec}}^{(1)}[\hat{\rho}_1], \quad (58)$$

$$\frac{d\hat{\rho}_0}{dt} = -\frac{i}{\hbar}[\hat{H}_0 + \hat{H}_{\text{int}}^{\text{eff}}(\hat{\rho}_1), \hat{\rho}_0] + \mathcal{L}_{\text{dec}}^{(0)}[\hat{\rho}_0], \quad (59)$$

with effective interactions

$$\hat{H}_{\text{int}}^{\text{eff}}(\hat{\rho}_0) = g(t) \left(\langle \hat{a}^\dagger \rangle_{\hat{\rho}_0} \hat{\sigma}_- + \langle \hat{a} \rangle_{\hat{\rho}_0} \hat{\sigma}_+ \right), \quad (60)$$

$$\hat{H}_{\text{int}}^{\text{eff}}(\hat{\rho}_1) = g(t) \left(\langle \hat{\sigma}_- \rangle_{\hat{\rho}_1} \hat{a}^\dagger + \langle \hat{\sigma}_+ \rangle_{\hat{\rho}_1} \hat{a} \right). \quad (61)$$

The mean-field approximation is valid when correlations between the two sectors are small, i.e.,

$$|\langle \hat{a} \hat{\sigma}_+ \rangle - \langle \hat{a} \rangle \langle \hat{\sigma}_+ \rangle| \ll |\langle \hat{a} \rangle \langle \hat{\sigma}_+ \rangle|. \quad (62)$$

For the numerical implementation, we do *not* use the mean-field approximation: we integrate the full master equation on the joint Hilbert space.

9 Observables

9.1 Analytic-mode population

The expectation of the analytic mode operator is

$$\boxed{\langle \hat{\sigma}_z \rangle(t) = \text{Tr}(\hat{\rho}(t) \hat{\sigma}_z)}. \quad (63)$$

The limits are

$$\langle \hat{\sigma}_z \rangle = +1 \iff \text{analytic mode active}, \quad \langle \hat{\sigma}_z \rangle = -1 \iff \text{analytic mode inactive}. \quad (64)$$

9.2 Substrate population

The mean number of substrate quanta is

$$\boxed{\langle \hat{n}_0 \rangle(t) = \text{Tr}(\hat{\rho}(t) \hat{a}^\dagger \hat{a})}. \quad (65)$$

9.3 Purity

The purity of the joint state is

$$\boxed{\mathcal{P}(t) = \text{Tr}(\hat{\rho}(t)^2)}. \quad (66)$$

For a pure state, $\mathcal{P} = 1$. For a maximally mixed state in dimension d , $\mathcal{P} = 1/d$.

9.4 von Neumann entropy

The entropy of the joint state is

$$\boxed{S_{vN}(t) = -\text{Tr}(\hat{\rho}(t) \ln \hat{\rho}(t)) = -\sum_i \lambda_i(t) \ln \lambda_i(t)}, \quad (67)$$

where λ_i are the eigenvalues of $\hat{\rho}$.

9.5 Steady-state averages

For a given parameter set, we run the simulation until $t_{\text{max}} = N_{\text{Rabi}} \cdot T_R$ and average over the final 10% of the trajectory:

$$\langle \hat{O} \rangle_{ss} = \frac{1}{N_{\text{avg}}} \sum_{i \in \text{final window}} \langle \hat{O} \rangle(t_i). \quad (68)$$

This removes transient oscillations.

10 Control parameters

10.1 Resonance parameter

We define the resonance parameter

$$\alpha \equiv \frac{g}{\delta}. \quad (69)$$

Regimes:

- $\alpha \ll 1$: dispersive regime (weak coupling).
- $\alpha \sim 1$: resonant regime (strong mixing).
- $\alpha \gg 1$: ultra-strong coupling (outside the RWA).

10.2 Dissipation ratio

We define the dissipation ratio

$$\mathcal{R} \equiv \frac{g}{\gamma_\phi + \gamma_a}. \quad (70)$$

Regimes:

- $\mathcal{R} \ll 1$: overdamped (dissipation dominates).
- $\mathcal{R} \sim 1$: critical.
- $\mathcal{R} \gg 1$: underdamped (coherent oscillations visible).

10.3 Combined parameter space

The dynamics depend on the pair $(\alpha, \mathcal{R})$:

$$\frac{\mathcal{R}}{\alpha} = \frac{\delta}{\gamma_\phi + \gamma_a} \equiv \mathcal{D}, \quad (71)$$

where $\mathcal{D}$ is a dimensionless "dissipation-detuning ratio". The phase diagram is effectively two-dimensional.

11 Information-theoretic measures

11.1 Channel capacity

The Shannon channel capacity for a phase-encoded signal is

$$C = B \log_2 (1 + \kappa), \quad (72)$$

where B is the channel bandwidth and

$$\kappa = \frac{g}{\gamma_\phi + \gamma_a} = \mathcal{R}. \quad (73)$$

For $\kappa \gg 1$: $C \approx B \log_2 \kappa$ (logarithmic growth). For $\kappa \ll 1$: $C \approx 1.44 B \kappa$ (linear growth).

11.2 Phase distribution

The stationary distribution of the phase difference $\Delta\phi = \phi_0 - \phi_1$ is the von Mises distribution

$$P(\Delta\phi) = \frac{e^{\kappa \cos(\Delta\phi - \phi_0)}}{2\pi I_0(\kappa)}, \quad (74)$$

where I_0 is the modified Bessel function of order zero. The concentration parameter κ is the same as above.

11.3 Phase entropy

The differential entropy of the phase distribution is

$$S_\phi(\kappa) = \ln(2\pi I_0(\kappa)) - \kappa \frac{I_1(\kappa)}{I_0(\kappa)}. \quad (75)$$

Limits:

$$\lim_{\kappa \rightarrow 0} S_\phi = \ln(2\pi) \approx 1.838 \text{ nats}, \quad \lim_{\kappa \rightarrow \infty} S_\phi = -\infty. \quad (76)$$

The relative entropy with respect to the uniform distribution is

$$D_{KL}(P||U) = \ln(2\pi) - S_\phi(\kappa) \geq 0. \quad (77)$$

11.4 Order parameter

The Kuramoto order parameter is

$$S(t) = \left| \langle e^{i\Delta\phi(t)} \rangle \right|. \quad (78)$$

For a perfectly phase-locked state, $S = 1$. For a uniformly distributed phase, $S = 0$.

12 Phase dynamics

12.1 Kuramoto–Adler equation

The phase difference evolves as

$$\frac{d\Delta\phi}{dt} = \Delta\omega - K \sin(\Delta\phi) + \xi(t), \quad (79)$$

where

- $\Delta\omega = \omega_1 - \omega_0 = \delta$.
- K : coupling strength (related to g).
- $\xi(t)$: noise with $\langle \xi(t)\xi(t') \rangle = 2D\delta(t - t')$.

12.2 Locking condition

The system is phase-locked when

$$K > |\Delta\omega|, \quad (80)$$

which corresponds in our model to

$$g > \delta \iff \alpha > 1. \quad (81)$$

This is a *different* threshold from $g_{\text{crit}} = \delta/2$. The Kuramoto locking condition gives $\alpha > 1$, while the geometric label $\theta = \pi/8$ gives $\alpha = 0.5$.

13 Numerical scheme

13.1 Discretization

We discretize the master equation as a linear system of ordinary differential equations. With $d = 2N_0$ the dimension of the Hilbert space, the density matrix has d^2 complex entries. Using Hermiticity and trace preservation, the effective number of real degrees of freedom is $d^2 - 1$. For $N_0 = 10$, this is 399.

13.2 Vectorization

We reshape $\hat{\rho}$ into a column vector

$$\boldsymbol{\rho} = \text{vec}(\hat{\rho}) \in \mathbb{C}^{d^2}. \quad (82)$$

The master equation becomes

$$\frac{d\boldsymbol{\rho}}{dt} = \mathcal{L}(t)\boldsymbol{\rho}, \quad (83)$$

where $\mathcal{L}(t)$ is the Liouvillian superoperator:

$$\mathcal{L}(t) = -i \left(\hat{H}(t) \otimes \hat{I} - \hat{I} \otimes \hat{H}(t)^T \right) + \mathcal{L}_{\text{dec}}. \quad (84)$$

13.3 Implicit integration

We use the backward differentiation formula (BDF) implicit method via `scipy.integrate.solve_ivp` with:

- Method: 'BDF'.
- Relative tolerance: $\text{rtol} = 10^{-7}$.
- Absolute tolerance: $\text{atol} = 10^{-9}$.

13.4 Why BDF?

The Liouvillian eigenvalues span a wide range:

$$\lambda_{\text{max}} \sim g\sqrt{N_0}, \quad \lambda_{\text{min}} \sim \gamma_\phi. \quad (85)$$

For $g = 50$ Hz, $N_0 = 10$, $\gamma_\phi = 0.05$ Hz:

$$\frac{\lambda_{\text{max}}}{\lambda_{\text{min}}} \sim 3 \times 10^4. \quad (86)$$

This stiffness ratio makes explicit methods (RK4, Euler) unstable. Implicit methods are required.

13.5 Hermiticity restoration

After each time step, numerical drift may break Hermiticity. We restore it by

$$\hat{\rho} \leftarrow \frac{1}{2} (\hat{\rho} + \hat{\rho}^\dagger). \quad (87)$$

Trace is restored by

$$\hat{\rho} \leftarrow \frac{\hat{\rho}}{\text{Tr}(\hat{\rho})}. \quad (88)$$

Table 1: Complete list of parameters used in Track A.

Symbol	Description	Value	Status
ω_0	Substrate frequency	6 Hz	free
ω_1	Analytic frequency	40 Hz	free
δ	Detuning	34 Hz	derived
N_0	Substrate truncation	10 (scan: 6)	numerical
g_0	Maximal coupling	20 Hz	free
λ	DMN sigmoid stiffness	8	free
A_{th}	DMN threshold	0.5	free
A_0	DMN baseline	0.3	free
τ_{DMN}	DMN relaxation time	2 s	free
σ	DMN noise amplitude	0.05	free
γ_ϕ	Phase damping rate	0.5 Hz	free
γ_a	Amplitude damping rate	0.1 Hz	free
n_{th}	Thermal occupation	0.05	free

14 Parameter table

15 Summary of key equations

For quick reference, the essential equations are:

Master equation.

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}(t), \hat{\rho}] + \mathcal{L}_{\text{dec}}[\hat{\rho}]. \quad (89)$$

Hamiltonian.

$$\hat{H}(t) = \omega_0 \hat{a}^\dagger \hat{a} + \omega_1 \hat{\sigma}_+ \hat{\sigma}_- + g(t) (\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+). \quad (90)$$

Dissipator.

$$\mathcal{L}_{\text{dec}}[\hat{\rho}] = \gamma_\phi (\hat{\sigma}_z \hat{\rho} \hat{\sigma}_z - \hat{\rho}) + \gamma_a (n_{\text{th}} + 1) \mathcal{D}[\hat{a}] \hat{\rho} + \gamma_a n_{\text{th}} \mathcal{D}[\hat{a}^\dagger] \hat{\rho}, \quad (91)$$

with $\mathcal{D}[\hat{L}] \hat{\rho} = \hat{L} \hat{\rho} \hat{L}^\dagger - \frac{1}{2} \{ \hat{L}^\dagger \hat{L}, \hat{\rho} \}$.

Coupling.

$$g(t) = g_0 \eta(A_{\text{DMN}}(t)), \quad \eta(A) = \frac{1}{1 + e^{\lambda(A - A_{\text{th}})}}. \quad (92)$$

Control parameters.

$$\alpha = \frac{g}{\delta}, \quad \mathcal{R} = \frac{g}{\gamma_\phi + \gamma_a}. \quad (93)$$

Observables.

$$\langle \hat{\sigma}_z \rangle = \text{Tr}(\hat{\rho} \hat{\sigma}_z), \quad \langle \hat{n}_0 \rangle = \text{Tr}(\hat{\rho} \hat{a}^\dagger \hat{a}). \quad (94)$$

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